

What is your counterparty's promise to pay worth?

CVA and DVA with simulation and *machine learning*

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2008: the lesson that changed derivative valuation

*During the financial crisis, close to two thirds of derivative losses came not from actual defaults, but from the **deterioration in the market value of the counterparties' promise to pay**.*

- A swap with a AAA bank and the same swap with a fragile counterparty **are not worth the same**. That discount has a name: **CVA**.
- It is now a **global requirement** —IFRS 13, Basel— and in Mexico a recurring regulatory deliverable to the CNBV and Banco de Mexico.
- That is why banks and corporates with derivatives run **dedicated XVA desks**: whole teams that compute and hedge this number. **Master this, and you work there.**

The whole idea in one formula

$$\text{CVA} = \underbrace{(1 - R)}_{\text{severity}} \int_0^T \underbrace{\text{EE}(t)}_{\text{exposure}} \underbrace{d\text{PD}(t)}_{\text{probability}}$$

Severity

If it defaults, how much is lost?

$R = 0.25$ (emerging markets)

Exposure

How much will it owe me at each future date?

We must simulate

Probability

How likely is it to default?
Read off the CDS market

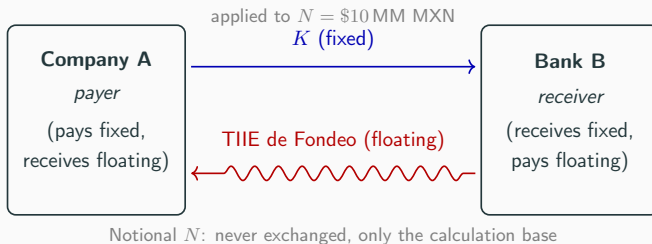
Why market probabilities and not historical ones? Because CVA is a **price** —hedged by buying CDS—, not an actuarial forecast: risk-neutral valuation, under the measure $\mathbb{Q}$.

The mirror image: DVA

$$\text{DVA} = (1 - R_{\text{own}}) \int_0^T \text{ENE}(t) d\text{PD}_{\text{own}}(t)$$

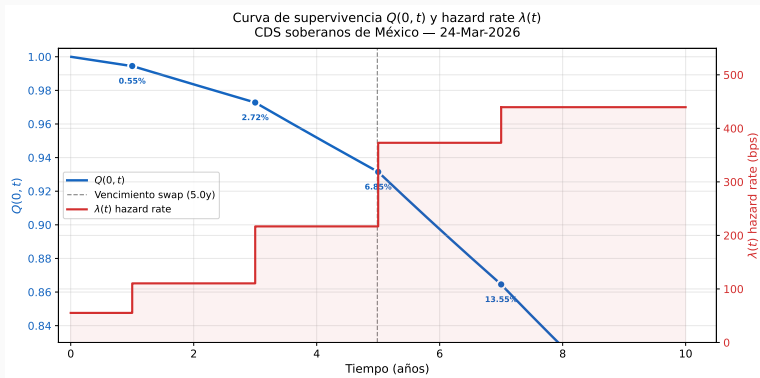
- The **same formula** seen from the other side of the table: the *negative* exposure with *our own* credit risk.
- Your counterparty is charging you a CVA right now; your DVA is that same number with the opposite sign.
- At the end of the class we compute it live, in three lines of code.

The instrument: an interest rate swap (IRS)



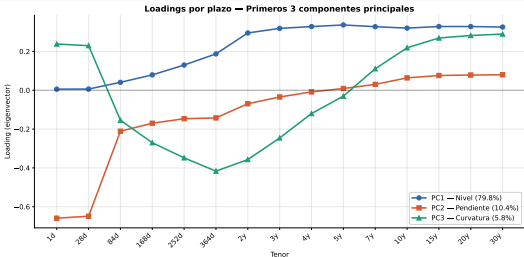
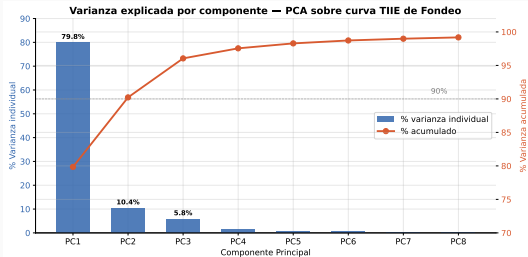
The *swap* in this class: fixed *payer*, 5 years, 10 MM MXN notional, TIIE de Fondeo, *at-the-money* — the live instrument in the Mexican market since the November 2024 migration.

Piece 1 — the probability is read off the market



- The CDS *spread* is the price of credit protection; a *bootstrap* turns it into a survival curve: $\lambda \approx s/(1 - R)$.

Piece 2 — exposure requires simulating the future

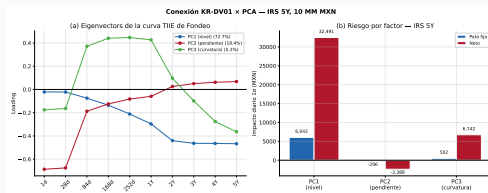


Level (79.85%) + slope (10.38%) = 90% of the curve. One factor is not enough; two are.

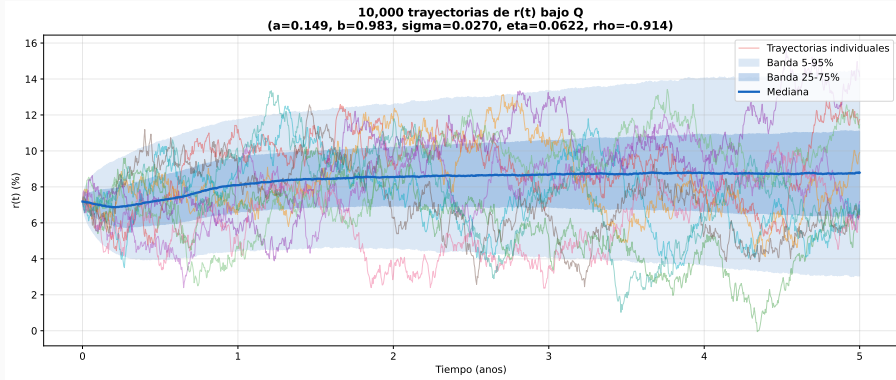
The model: G2++, two correlated Gaussian factors

$$r(t) = x(t) + y(t) + \varphi(t), \quad \begin{aligned} dx(t) &= -a x(t) dt + \sigma dW_1(t) \\ dy(t) &= -b y(t) dt + \eta dW_2(t) \end{aligned} \quad dW_1 dW_2 = \rho dt$$

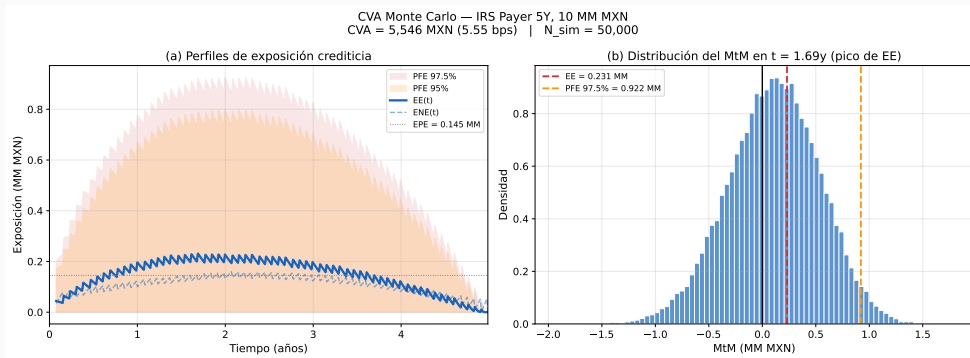
- **Slow** factor ($a = 0.149$): persists $\rightarrow$ **level** (PC1).
- **Fast** factor ($b = 0.983$): dies out $\rightarrow$ **slope** (PC2).
- $\rho = -0.914$: together they reproduce the shapes the PCA observes.
- Calibrated to ATM *swaptions* (Valmer); **exact** simulation.



10,000 possible futures of TIIE de Fondeo



The signature of CVA: the exposure hump



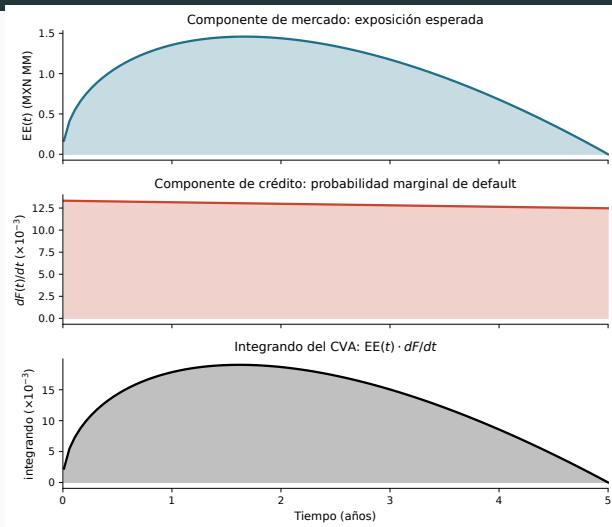
It rises through rate diffusion and falls through amortisation (*pull-to-par*); the peak arrives at $t \approx 1.69$ years ($\approx T/3$).

The price of counterparty risk

$$\text{CVA} = (1 - R) \int_0^T \text{EE}(t) d\text{PD}(t)$$

$$= 5,546 \text{ MXN} = 5.55 \text{ bps}$$

The hump $\times$ the probability: the integrand tells us *when* the risk bites. That is the discount the desk builds into the quote, today.



From one swap to the book: the operational problem

A 264 ms calculation sounds fast. . . until you multiply it:

- A book of 795 *swaps* $\Rightarrow$ ~ 3.5 minutes per full valuation.
- $\times 40$ bumps for sensitivities $\Rightarrow \sim 2.3$ hours.
- $\times 1,000$ stress scenarios $\Rightarrow$ days of compute.

And the desk does not only file the report: **it needs answers during the day.**

Periodic (regulatory)

- NIF C-10 / Annex 33 CUB (CNBV)
- Banxico XVA report: CVA counterparty by counterparty

Intraday (operational)

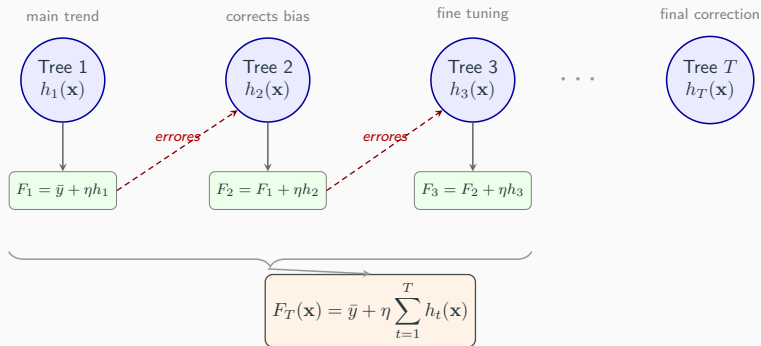
- Quoting: how much CVA does *this* trade add?
- Sensitivities and hedging the book
- Where does the CVA come from?
Trade-level attribution

Machine learning: learning from the engine itself

1. The Monte Carlo engine generates thousands of examples (trade, market) → CVA.
The labels are free: the teacher is our own simulator.
2. A supervised model learns that function. Two families, two different ideas: **trees that correct each other** (XGBoost) and **layers that compose** (neural network).
3. The surrogate answers in **microseconds**; the engine still signs the auditable report.

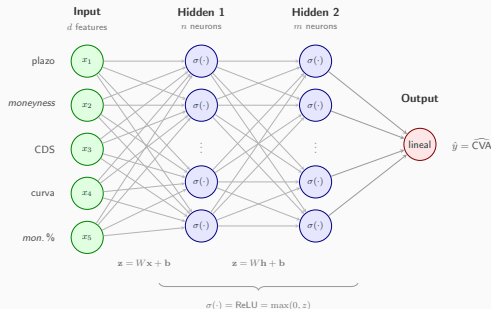
We are not replacing the engine: we clone it for the uses where its cost gets in the way.

XGBoost: hundreds of trees correcting one another



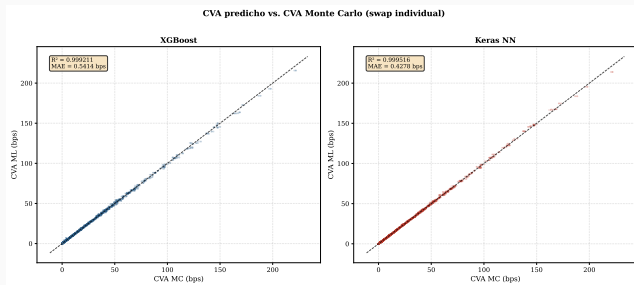
Each tree corrects the errors of the previous one (*boosting*); the ensemble learns CVA to within fractions of a basis point.

The neural network: a composition of simple functions



It is trained by adjusting weights until the validation error stops falling; exact gradients come from automatic differentiation.

How well does it learn? $R^2 = 0.999$



	Monte Carlo	XGBoost
Time per swap	264 ms	$1.7 \mu s$
Speed-up	$1\times$	$\sim 155,000\times$

This is *query latency*: the engine still generates the labels, validates the surrogate and keeps the auditable report.

In the *notebook*, in this order:

1. Simulate 10,000 G2++ paths (new seed, new future).
2. Watch the hump and the CVA of the reference *swap* appear.
3. **DVA in three lines**: the mirror image, with our own *spread*.
4. Train the surrogate and run the race: milliseconds vs. microseconds.

Everything you have just seen is taught here

- Stochastic processes and the measure $\mathbb{Q}$ → the G2++ and the simulation.
- Numerical methods → Monte Carlo, *bootstrap*, calibration.
- *Machine learning* → the surrogate pricer and its validation.
- Credit and counterparty risk → CVA, DVA, exposure.
- Intraday portfolio management → *netting*, incremental *pre-deal* CVA, sensitivities and trade-level attribution.
- Mexican regulation → CNBV, Banco de Mexico, the XVA report.

The master's programme is the bridge between the formula and the desk.



The complete *notebook* for this class:

- G2++ simulation and *swap* valuation
- CVA, DVA and bilateral netting
- The *machine learning* surrogate

Runs in the browser with Google Colab —nothing to install— or on your machine with `pip install -r requirements.txt`.

Questions?

Market data: Valmer